Science Week 3 plus: forces (written)

Ember · Real AQA GCSE Physics questions · about 15 minutes · pencil and ruler Date: _____

Skill: the Because Chain. One mark per link. Plan the chain in the boxes first, then write it as joined-up sentences using *as, until, so, which means*.

The context for both questions. In a world record attempt, a skydiver jumped from a height of 40 km above the Earth.

AQA Physics 8463/2H, June 2025, Q05.1 **[2 marks]**

(Draw your answer on Figure 6.)

Figure 6



June 2025, Paper 2, Figure 6

Figure 6 shows an incomplete free-body diagram for the skydiver a few seconds after the start of the jump. The skydiver has not reached terminal velocity. Complete the free-body diagram in Figure 6.

Plan the chain for the next question (one short phrase per box)

Cause	Effect	Until...	So...	So...

AQA Physics 8463/2H, June 2025, Q05.3 [4 marks]

The skydiver accelerated until reaching terminal velocity. Explain why the skydiver reached terminal velocity.

Self-check (5 min)

□ My free-body arrow starts at the dot, points up, and is shorter than the weight arrow.

□ I labelled it.

□ My explanation has four links, each joined to the last.

□ I said air resistance becomes **equal** to the weight.

□ I finished with what happens to the acceleration.

AQA mark scheme: Q05.1 (2 marks)

- a single arrow starting from the dot, pointing vertically upwards **and** shorter than the weight arrow **[1]**
- labelled air resistance (allow drag) **[1]**

Why shorter? The skydiver has not reached terminal velocity yet, so the forces are not balanced: weight is still bigger.

AQA mark scheme: Q05.3 (4 marks)

- (as the skydiver's velocity / speed increased) the air resistance increased **[1]**
- until the air resistance was equal to the weight of the skydiver (and in the opposite direction) **[1]**
- (so) the resultant force was zero **[1]**
- (so) the acceleration was zero (so the velocity was constant) **[1]**

Allow "drag" for air resistance throughout. Score yourself: ____ / 4. Which link did you miss? Write it in now, in a different colour.

Questions and mark scheme wording © AQA, from published GCSE Physics 8463 papers (June 2025). Used for private practice.